

L-15

Chapter - 6.1, 6.2

Least Square Approximation

→ if number of equation $>$ number of variables,
then Gaussian Elimination / LU decomposition
can't be used!

→ Gaussian Elimination / LU decomposition is
applicable only for Square Matrix.

→ In this topic we learn, to convert
a Non-square (Overdetermined) matrix to a
Square Matrix.

Ex.

$$2x + 5y + 2z = 10$$

$$5x + 2y + z = 20$$

$$2x + 2y + 5z = 30$$

$$x + 3y + 4z = 25$$

coefficient
Matrix

$$\begin{bmatrix} 2 & 5 & 2 \\ 5 & 2 & 1 \\ 2 & 2 & 5 \\ 1 & 3 & 4 \end{bmatrix}_{4 \times 3}$$

$m \times n$ size.
where, $m > n$

Not a Square Matrix
It's a Overdetermined Matrix
($m \times n$)

So, we can't apply : $AX = B$ (only for square matrix)

We have to do:

$$A^T \cdot A \cdot X = A^T \cdot B$$

why?

A size = $m \times n$

A^T size = $n \times m$

So, $A^T \cdot A = [n \times m] \cdot [m \times n]$

Square $\rightarrow [n \times n]$ size

Ex. Say, a 2 degree polynomial function.

x	$f(x)$
2	10
4	15
5	20
10	50

Ans:

So

$$P_2(x) = a_0 + a_1x + a_2x^2$$

$$\begin{cases} \therefore P_2(2) \Rightarrow a_0 + 2a_1 + 4a_2 = 10 \\ \therefore P_2(4) \Rightarrow a_0 + 4a_1 + 16a_2 = 15 \\ \therefore P_2(5) \Rightarrow a_0 + 5a_1 + 25a_2 = 20 \\ \therefore P_2(10) \Rightarrow a_0 + 10a_1 + 100a_2 = 50 \end{cases}$$

→ Matrix:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 10 & 100 \end{bmatrix}_{4 \times 3}; X = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}; B = \begin{bmatrix} 10 \\ 15 \\ 20 \\ 50 \end{bmatrix}$$

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not a square matrix!

Need: $A^T A \cdot X = A^T \cdot B$

3×4 $\rightarrow$ 4×3

Here $A^T \cdot A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 5 & 10 \\ 4 & 16 & 25 & 100 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 10 & 100 \end{bmatrix}$

$$= \begin{bmatrix} 4 & 21 & 145 \\ 21 & 145 & 1157 \\ 145 & 1157 & 9809 \end{bmatrix}_{3 \times 3}$$

$$A^T \cdot B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 5 & 10 \\ 4 & 16 & 25 & 100 \end{bmatrix} \cdot \begin{bmatrix} 10 \\ 15 \\ 20 \\ 50 \end{bmatrix}$$

$$= \begin{bmatrix} 95 \\ 730 \\ 6350 \end{bmatrix}$$

We get: $A^T A X = A^T \cdot B$

Using Gaussian Elimination to Solve!

$$\Rightarrow \begin{bmatrix} 4 & 21 & 145 \\ 21 & 145 & 1157 \\ 145 & 1157 & 9809 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 95 \\ 730 \\ 6350 \end{bmatrix}$$

Augmented Matrix $\rightarrow \left[\begin{array}{ccc|c} 4 & 21 & 145 & 95 \\ 21 & 145 & 1157 & 730 \\ 145 & 1157 & 9809 & 6350 \end{array} \right]$

Step 1
doing: $R_2 = R_2 - \left(\frac{21}{4}\right) R_1$
 $R_3 = R_3 - \left(\frac{145}{4}\right) R_1$

we get,

$$\left[\begin{array}{ccc|c} 4 & 21 & 145 & 95 \\ 0 & 34.75 & 395.75 & 231.25 \\ 0 & 395.75 & 4552.75 & 2906.25 \end{array} \right]$$

Step 2: $R_3 = R_3 - \left(\frac{395.75}{34.75}\right) R_2$

we get

$$\left[\begin{array}{ccc|c} 4 & 21 & 145 & 95 \\ 0 & 34.75 & 395.75 & 231.25 \\ 0 & 0 & 41.2 & 270 \end{array} \right]$$

Now

$$\left[\begin{array}{ccc|c} 4 & 21 & 145 & 95 \\ 0 & 34.75 & 395.75 & 231.25 \\ 0 & 0 & 41.2 & 270 \end{array} \right] \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 95 \\ 231.25 \\ 270 \end{bmatrix}$$

$$41.2 a_2 = 270 \Rightarrow a_2 = 0.5$$

$$0 \cdot a_0 + 34.75 a_1 + 395.75 a_2 = 231.25 \Rightarrow a_1 = -1$$

$$4 a_0 + 21 a_1 + 145 a_2 = 95 \Rightarrow a_0 = 5$$

So, $\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 0.5 \end{bmatrix}$

Aus

□ Orthogonal = Orthogonal + Normal

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Basis are 90° of each others.



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if this satisfies then they are Orthogonal

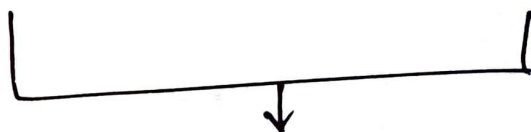
$$\begin{cases} X^T \cdot Y = 0 \\ Y^T \cdot Z = 0 \\ X^T \cdot Z = 0 \end{cases}$$

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Transpose & or matrix gets constant unit (1) value.

→
(1)

⇓

$$\begin{cases} X^T \cdot X = 1 \\ Y^T \cdot Y = 1 \\ Z^T \cdot Z = 1 \end{cases} \left\{ \begin{array}{l} \text{if this} \\ \text{satisfy} \\ \text{then the} \\ \text{basis} \\ \text{is} \\ \text{Normal.} \end{array} \right.$$



if both satisfies
then → Orthogonal

Ex. $X = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} ; Y = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$

Prove they're orthogonal & normal. [Orthonormal]

$$\Rightarrow X^T \cdot Y = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix} = 0$$

So, orthogonal.

$$X^T \cdot X = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = 1$$

So, X is normal

$$Y^T \cdot Y = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix} = 1$$

Y is normal.

As both are normal & orthogonal,

So, they're orthonormal.

Q How orthonormal is linked with Kronecker Delta?

$$\delta_{ij} = \begin{cases} 1 & i=j \longrightarrow x^T \cdot x = \textcircled{1} \text{ [Normal]} \\ & \hookrightarrow i, j \text{ if same} \\ 0 & i \neq j \longrightarrow x^T \cdot y = \boxed{0} \text{ [Orthogonal]} \\ & \hookrightarrow i, j \text{ different matrix} \end{cases}$$